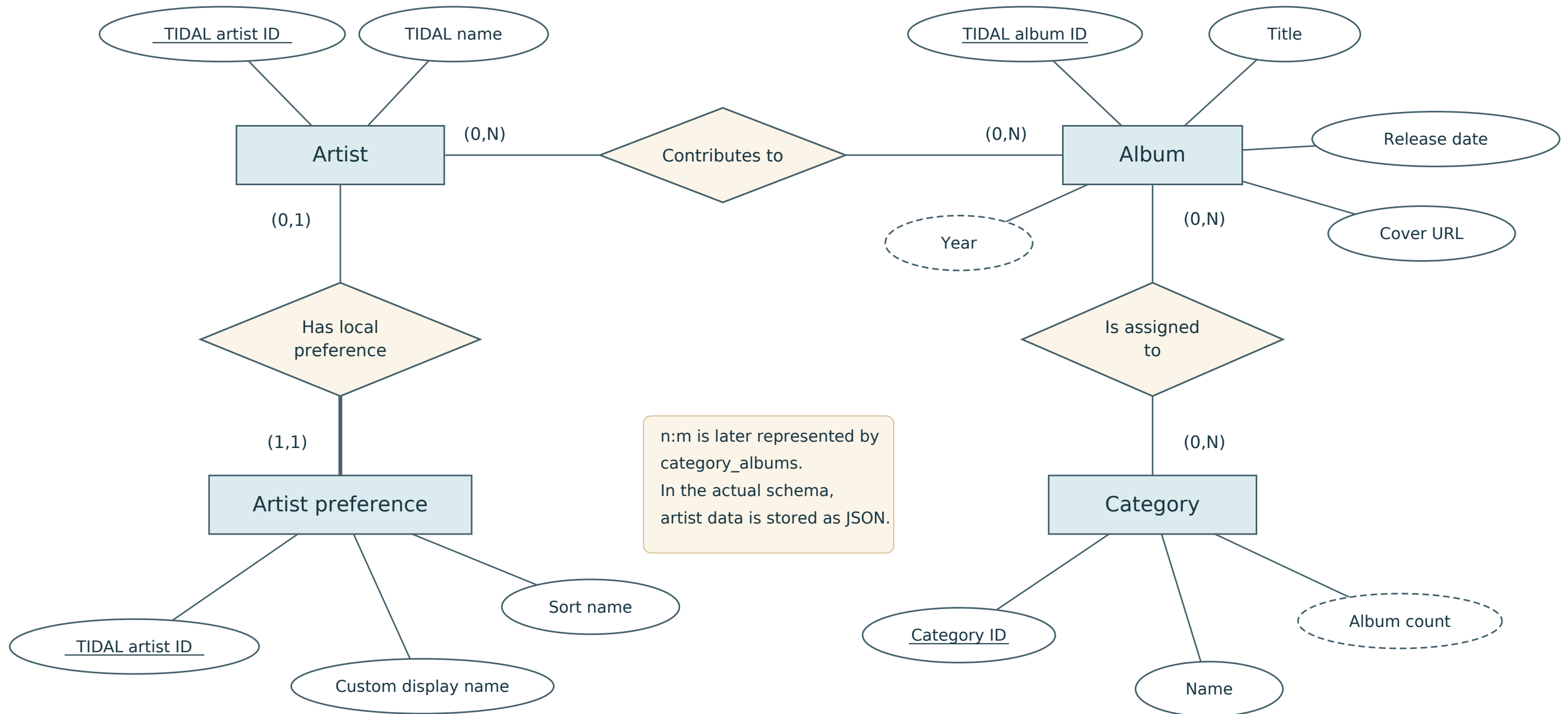


ER model in Chen notation

Entities, attributes and relationships - not yet mapped to SQL tables.



Rectangle: entity · Diamond: relationship · Ellipse: attribute · Underlined: key · Dashed ellipse: derived.

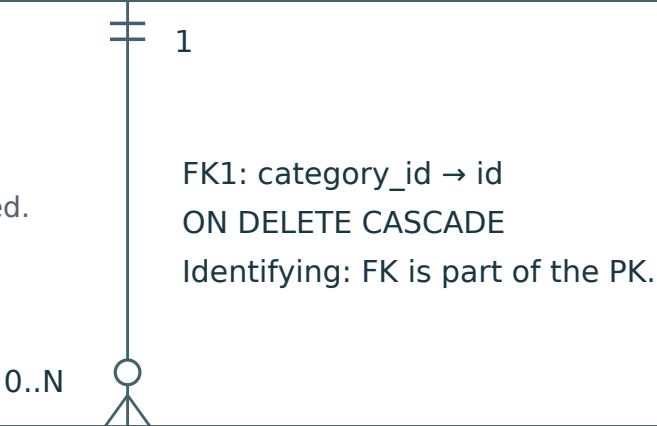
(min,max) applies to each instance of the adjacent entity. 0 = optional; N = any number. Thick line = total participation.

Album → Artist: minimum 0 because the app allows missing metadata. Technical cache fields and timestamps are in the table model.

Actual schema: tidal2wiim.db, version 3 · Only actually declared SQL foreign keys have relationship lines.

categories			
Attribute	Data type	NULL	Markers
<u>id</u>	INTEGER	no ¹	PK / AUTO
name	TEXT	no	UQ
created_at	INTEGER	no	
updated_at	INTEGER	no	

name: UNIQUE + NOCASE.
¹ id: INTEGER PRIMARY KEY
with AUTOINCREMENT.
SQLite generates the ID if needed.



category_albums			
Attribute	Data type	NULL	Markers
<u>category_id</u>	INTEGER	no	PK / FK1
<u>album_id</u>	TEXT	no	PK

PK = (category_id, album_id), not two separate primary keys.
album_id is NOT an SQL FK to album_cache.
An album can remain assigned even if its cache row is missing.
Index: idx_category_albums_album(album_id).
FK1 is enabled by PRAGMA foreign_keys = ON.

album_cache			
Attribute	Data type	NULL	Markers
<u>album_id</u>	TEXT	no	PK
collection_position	INTEGER	no	
title	TEXT	no	
artists_json	TEXT	no	
release_date	TEXT	yes	
cover_url	TEXT	yes	
metadata_missing	INTEGER	no	Default 0
cached_at	INTEGER	no	

artists_json: JSON list of artist ID and name; no artist table.
Index: idx_album_cache_position(collection_position).
metadata_missing: app writes 0/1, but SQL has no CHECK.
release_date: ISO 8601 text; cached_at: Unix time in milliseconds.

artist_preferences			
Attribute	Data type	NULL	Markers
<u>artist_id</u>	TEXT	no	PK
custom_name	TEXT	yes	
sort_name	TEXT	no	
updated_at	INTEGER	no	

artist_id is a TIDAL reference, but not an SQL foreign key.
Preferences exist independently of the current album cache.

PK = primary key · FK1 = declared foreign key · UQ = UNIQUE · AUTO = AUTOINCREMENT · Timestamps as INTEGER (ms).